

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

M. Tech. (CE) Sem. – II University Examination, 2010-11

CE710 High Performance Computing

Date: 16.05.2011, Monday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper.

SECTION-I

Q-1

- | | |
|--|---|
| (a) Explain data access modes in ViPIOS. | 3 |
| (b) Does Open MPI support checkpoint and restart of parallel jobs? | 3 |
| (c) Why do we need metacomputing? Explain with an example. | 3 |
| (d) What are the limitations to the pipelining? | 2 |

Q-2 Answer the following questions.

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|---|---|
| (a) Draw and explain cluster computer architecture. | 5 |
| (b) Write a short note on PLUS architecture. | 4 |
| (c) Differentiate vector and parallel processing | 3 |

OR

- | | |
|--|---|
| (b) What is supercomputer and how do they differ from a personal computer? | 4 |
| (c) How to compute CPU time per workload type? | 3 |

Q-3 Answer the following questions. (Any Three)

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|---|----|
| (a) Explain Cray 2 supercomputer. | 12 |
| (b) What are the methods to provide fault tolerance in distributed systems? Explain in detail. | |
| (c) What is high-performance cluster? Explain in terms of computing. | |
| (d) Consider a two-level memory hierarchy, M_1 and M_2 . Denote the hit ratio of M_1 as h . let c_1 and c_2 be costs per KB, s_1 and s_2 be the memory capacities, and t_1 and t_2 be the access times, respectively.
(a) Under what condition will the average cost of the entire memory system approach the value of c_2 ?
(b) What is the effective memory-access time, t_a , of this hierarchy? | |

SECTION-II

Q-4 Answer the following questions.

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|---|---|
| (a) What are the applications of Network RAM? | 2 |
| (b) What are the distinguish characteristics of PARMA project? | 2 |
| (c) What is granularity in Distributed Shared Memory? | 2 |
| (d) What is Double-faulting problem in Distributed Shared memory? What are the mechanisms of solution of it? | 2 |
| (e) What is major problem associated in start and tree topology network? How Fat tree can solve that problem? | 3 |

Q-5 Answer the following questions.

- (a) What is Relaxing Consistency? Write down the various mechanism of it. 5
- (b) Write down various techniques to Optimize Communication in inter-process communication. What is the importance of Efficient Collective Communications? 7

OR

- (b) In Network RAM, failure of workstation and network causes unavailability pages. What are the solutions to overcome that situation? Which one is very efficient among those solutions? 7

Q-6 Answer the following questions.(Any Three)

12

- (a) How barrel shifter can be made? Draw barrel shifter with network size=8. Write down degree and diameter of barrel shifter with respect to network size=8.
- (b) What are the pros and cons of GMS and Remote memory paging mechanism?
- (c) What are the two approaches of write synchronization problem in Distributed Shared memory?
- (d) Write a short note on: HiPPI.